

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

ID: SOLUTION

Name: SOLUTION

Section:

1. Consider a unity feedback closed loop system with its open loop transfer function as $G(s)$. A portion of the Nyquist plot of $G(s)$ is shown in Fig. Q1 (solid line), along with a unit circle (dotted line). The plot intersects the real axis (at **Point 'A'**) for a frequency of $\omega = 10.2$ rad/s and intersects the unit circle (at **Point 'B'**) for a frequency of $\omega = 7.72$ rad/s.

- Find the gain margin (in dB).
- Find the phase margin (in degree).
- If the open loop gain of this system is increased by a factor of 1.709, what will be the new phase crossover frequency?

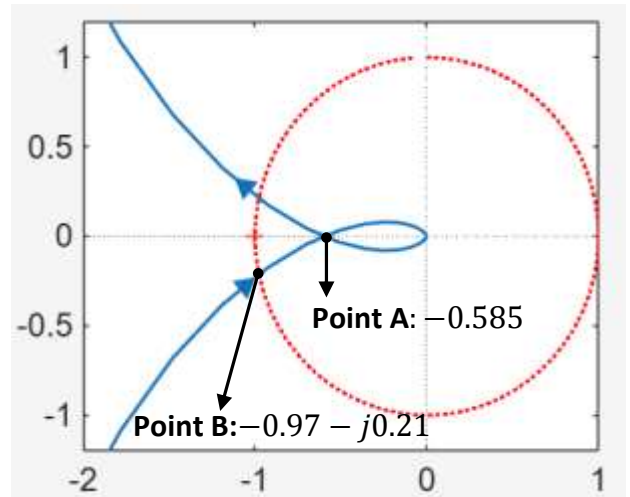


Fig. Q1

A) Gain Margin = $20 \log(1/0.585) = 4.65$ dB

B) P.M. = $180 + \angle(-0.97 - j0.21) = 180 - 167.8 = 12.2$ degree

C) If the gain increases by 1.709 times, the real axis intersection will be at $-0.585 \times 1.709 = (-1 + j0)$ point. So new phase crossover frequency will be 10.2 rad/s.

Alternatively, gain variation does not cause any change in phase variation, so phase crossover frequency will not change and remain same as 10.2 as observed from Fig. Q1.

A. Gain Margin (in dB) = 4.65 dB 2 M

B. Phase Margin (in degrees) = 12.2 2M

C. New phase crossover frequency (in rad/s) = 10.2 rad/s 2M